

Jiarui Liu

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🌐 Homepage | 🎓 Google Scholar | 🐦 Twitter | 🔗 LinkedIn | 🐙 GitHub

Research interests: reasoning, continual learning, reinforcement learning, agents, alignment.

EDUCATION

Carnegie Mellon University

Ph.D. in Language Technologies, SCS, LTI; GPA: 4.00/4.00

• **Advisor:** Mona Diab, department head of LTI at CMU

Pittsburgh, Pennsylvania, US

Aug. 2025 – May 2028 (Expected)

Carnegie Mellon University

Master of Language Technologies (MLT), SCS, LTI; GPA: 4.00/4.00

• **Advisor:** Mona Diab, department head of LTI at CMU

• **Relevant coursework:** LLM Systems (A+), Neural Code Generation (A+), NLP Ethics (A), Advanced NLP (A), Multimodal ML (A), Speech Recognition (A), Quantitative Evaluation

Pittsburgh, Pennsylvania, US

Aug. 2023 – May 2025

University of Michigan, Ann Arbor

BSE in Computer Science (Dual Degree); GPA: 3.86/4.00

• **Relevant coursework:** Intro to NLP (A), Science in Deep Learning (A), Operating Systems, Machine Learning (A), Deep Learning in CV (A+), Data Structures and Algorithms (A), Practical Programming in Java (A+), Computer Architecture (A), Web Systems (A-), Mobile App Development (A)

Ann Arbor, Michigan, US

Sep. 2021 – May 2023

Shanghai Jiao Tong University Joint Institute (UM-SJTU JI)

BSE in ECE (Dual Degree);

• **Relevant coursework:** Programming and Elementary Data Structures, Probability Methods and Statistics, Linear Algebra, Discrete Mathematics, Honor Mathematics

Shanghai, China

Sep. 2019 – Aug. 2023

PUBLICATIONS & PREPRINTS

* indicates equal contribution. † indicates equal supervision.

37. "Learning from Use: Test-Time Learning in Large Language Models and Agents"

Weihao Xuan*, Qingcheng Zeng*, Jiarui Liu*, Rui Yang*, Yunze Xiao, Yinxi Li, Zeqi Zhou, Weiwei Sun, Heli Qi, Leyang Cui, Tianhao Ma, Hanqun Cao, Junjue Wang, Tianyu Liu, Weihua Du, Yueqi Song, Jiayi Geng, Matthew Yu Heng Wong, Rob Voigt, Chuan Hong, Xiang Yue, Jen-Tse Huang, Xuhai Xu, Yifan Peng, Pheng Ann Heng, Yiming Yang, Bernhard Schölkopf, Zhijing Jin, Nan Liu, Naoto Yokoya, and Mona Diab

[SSRN]

36. "Evaluating Multi-Turn Multimodal Diagnostic Reasoning on Challenging Real-World Clinical Cases"

Rui Yang, Weihao Xuan, Yi Lin, Zhuhan Bao, Jonathan Chong Kai Liew, Matthew Yu Heng Wong, Nicolás Lescano, Nikita R Paripati, Emily Ling-Lin Pai, **Jiarui Liu**, Heli Qi, Heng-Jui Chang, Benny Kai Guo Loo, Huitao Li, Kunyu Yu, Yufan Wang, Chuan Hong, Shijian Lu, Douglas Teodoro, Naoto Yokoya, Ross Koppel, Mona Diab, Hua Xu, David W Bates, Nan Liu, Yifan Peng

[Arxiv]

35. "PACE: A Proxy for Agentic Capability Evaluation"

Yueqi Song, Lintang Sutawika, **Jiarui Liu**, Lindia Tjuaatja, Jiayi Geng, Yunze Xiao, Daniel Lee, Aditya Bharat Soni, Vincent Lo, Xiang Yue, Graham Neubig

[Arxiv]

34. "PaperMentor: A human-centered multi-agent writing tutor for AI research papers in Overleaf"

Jiarui Liu, Terry Jingchen Zhang, Ryan Faulkner, Xuanqiang Angelo Huang, Vilém Zouhar, Dominik Glandorf, Isabel Dahlgren, Rishit Dagli, Yuen Chen, Felix Leeb, Van Q Truong, Punya Syon

Pandey, Yves Bicker, Suvajit Majumder, Wenyan Jiang, Zeju Qiu, Sankalan Pal Chowdhury, Mrinmaya Sachan, Bernhard Schölkopf, Mona Diab, Zhijing Jin

ACL 2026 Demo [\[ACL Anthology\]](#) [\[Code\]](#) [\[Demo\]](#)

33. *"OdysSim: Building Foundation Models for Human Behavior Simulation"*

Xuhui Zhou, Weiwei Sun, Weihua Du, **Jiarui Liu**, Haojia Sun, Qianou Ma, Tongshuang Wu, Yiming Yang, Maarten Sap

[\[Arxiv\]](#)

32. *"Re-Centering Humans in LLM Personalization"*

Lechen Zhang, **Jiarui Liu**, Tal August

[\[Arxiv\]](#)

31. *"Knowledge Index of Noah's Ark"*

Sheng Jin, Minghao Liu, Yunze Xiao, Zeqi Zhou, Heli Qi, Yifan Yao, Meishu Song, Kaijing Ma, Xuan Zhang, Sicong Jiang, Yizhe Li, Ningshan Ma, Jie Wei, Ziniu Li, Minglai Yang, Bangya Liu, Yiming Liang, Xiao Fang, Qingcheng Zeng, **Jiarui Liu**, Rui Yang, Shen Yan, Wenhao Huang, Jiaheng Liu, Zihan Wang, Weihao Xuan, Ge Zhang

[\[Arxiv\]](#)

30. *"LT2: Linear-Time Looped Transformers"*

Chunyan Deng, Yizhe Zhang, Rui-Jie Zhu, Yuanyuan Xu, **Jiarui Liu**, TS Ng, Hanjie Chen

[\[Arxiv\]](#)

29. *"Reinforcing Human Behavior Simulation via Verbal Feedback"*

Weiwei Sun, Xuhui Zhou, **Jiarui Liu**, Weihua Du, Haojia Sun, Yiqing Xie, Qianou Ma, Sihao Chen, Mengting Wan, Longqi Yang, Pei Zhou, Sherry Wu, Sean Welleck, Graham Neubig, Yiming Yang, Maarten Sap

[\[Arxiv\]](#)

28. *"MixSD: Mixed Contextual Self-Distillation for Knowledge Injection"*

Jiarui Liu, Lechen Zhang, Yongjin Yang, Yinghui He, Yingheng Wang, Weihao Xuan, Zhijing Jin, Mona Diab

[\[Arxiv\]](#)

27. *"Transferability for General Reasoning: An Automated Curriculum for Multi-Domain RLVR"*

Yongjin Yang, **Jiarui Liu**, Yinghui He, Lezhen Zhang, Bernhard Schölkopf, Zhijing Jin

Best Paper Award at ICML RLxF Reinforcement Learning from World Feedback Workshop 2026

[\[Arxiv\]](#)

26. *"Self-distillation zero: Self-revision turns binary rewards into dense supervision"*

Yinghui He, Simran Kaur, Adithya Bhaskar, Yongjin Yang, **Jiarui Liu**, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, Sanjeev Arora

Best Paper Award at ICML RLxF Reinforcement Learning from World Feedback Workshop

[\[Arxiv\]](#)

25. *"CLT-Forge: A Scalable Library for Cross-Layer Transcoders and Attribution Graphs"*

Florent Draye, Abir Harrasse, Vedant Palit, Tung-Yu Wu, **Jiarui Liu**, Punya Syon Pandey, Roderick Wu, Terry Jingchen Zhang, Zhijing Jin, Bernhard Schölkopf

[\[Arxiv\]](#)

24. *"Mind the sim2real gap in user simulation for agentic tasks"*

Xuhui Zhou, Weiwei Sun, Qianou Ma, Yiqing Xie, **Jiarui Liu**, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, Maarten Sap

[\[Arxiv\]](#)

23. *"Making Complex Reasoning Student-Friendly: A Hybrid LLM-to-SLM Distillation Framework"*

Yongjin Yang, Yinghui He, **Jiarui Liu**, Zhijing Jin

Last updated in Jan. 2026

22. *"Stabilizing Reinforcement Learning for Honesty Alignment in Language Models on Deductive Reasoning"*
Jiarui Liu, Kaustubh Dhole, Yingheng Wang, Haoyang Wen, Sarah Zhang, Haitao Mao, Gaotang Li, Neeraj Varshney, Jingguo Liu, Xiaoman Pan
AAAI 2026 Bridge LMReasoning Workshop. AAAI 2026 MATH4AI Workshop. [\[Arxiv\]](#)
21. *"LLM Microscope: What Model Internals Reveal About Answer Correctness and Context Use"*
Jiarui Liu*, Jivitesh Jain*, Mona Diab, Nishant Subramani
COLM 2025 Interplay Workshop. [\[Arxiv\]](#)
20. *"CauSciBench: A Comprehensive Benchmark on End-to-End Causal Inference for Scientific Research"*
Sawal Acharya, Terry Jingchen Zhang, Andrew Kim, Anahita Haghighat, Sun Xianlin, Pepijn Cobben, Rahul Babu Shrestha, Maximilian Mordig, Jacob T. Emmerson, Furkan Danisman, Yuen Chen, Clijo Jose, Andrei Ioan Muresanu, Justin Cui, **Jiarui Liu**, Yahang Qi, Punya Syon Pandey, Yinya Huang, Bernhard Schölkopf, Mrinmaya Sachan
ICML 2026.
19. *"Taming Object Hallucinations with Verified Atomic Confidence Estimation"*
Jiarui Liu, Weihao Xuan, Zhijing Jin, Mona Diab
EACL 2026. [\[Arxiv\]](#)
18. *"CORE: Measuring Multi-Agent LLM Interaction Quality under Game-Theoretic Pressures"*
Punya Syon Pandey, Yongjin Yang, **Jiarui Liu**, Zhijing Jin
EACL 2026. [\[Arxiv\]](#)
17. *"Synthetic Socratic Debates: Examining Persona Effects on Moral Decision and Persuasion Dynamics"*
Jiarui Liu, Yueqi Song*, Yunze Xiao*, Mingqian Zheng*, Lindia Tjauatja, Jana Schaich Borg, Mona Diab, Maarten Sap
EMNLP 2025 Main. [\[Arxiv\]](#)
16. *"Humanizing Machines: Rethinking LLM Anthropomorphism Through a Multi-Level Framework of Design"*
Yunze Xiao*, Lynnette Hui Xian Ng*, **Jiarui Liu**, Mona Diab
EMNLP 2025 Main Oral. [\[Arxiv\]](#)
15. *"BIG5-CHAT: Shaping LLM Personalities Through Training on Human-Grounded Data"*
Wenkai Li*, **Jiarui Liu***, Andy Liu, Xuhui Zhou, Mona T. Diab, Maarten Sap
ACL 2025 Main. [\[Arxiv\]](#) [\[Code\]](#)
14. *"Towards Global AI Inclusivity: A Large-Scale Multilingual Terminology Dataset (GIST)"*
Jiarui Liu*, Iman Ouzzani*, Wenkai Li*, Lechen Zhang, Tianyue Ou, Houda Bouamor, Zhijing Jin, Mona Diab
ACL 2025 Findings. [\[Arxiv\]](#)
13. *"Uncovering and Understanding Social Media Censorship across Countries"*
Neemesh Yadav*, **Jiarui Liu***, Francesco Ortu, Zhijing Jin, Rada Mihalcea
ACL 2025 Findings.
12. *"Chumor 2.0: Towards Benchmarking Chinese Humor Understanding"*
Ruiqi He, Yushu He, Longju Bai, **Jiarui Liu**, Zhenjie Sun, Zenghao Tang, He Wang, Hanchen Xia, Rada Mihalcea, Naihao Deng
ACL 2025 Findings. [\[Arxiv\]](#)
11. *"EmoNews: A Spoken Dialogue System for Expressive News Conversations"*
Ryuki Matsuura*, Shikhar Bharadwaj*, **Jiarui Liu***, Dhatchi Kunde Govindarajan
SigDial 2025 Demo. [\[Arxiv\]](#)
10. *"Language Model Alignment in Multilingual Trolley Problems"*
Zhijing Jin*, Max Kleiman-Weiner*, Giorgio Piatti*, Sydney Levine, **Jiarui Liu**, Fernando Gon-

zalez, Francesco Ortu, András Strausz, Mrinmaya Sachan, Rada Mihalcea, Yejin Choi, Bernhard Schölkopf

ICLR 2025 and Best Paper Award at NeurIPS Pluralistic Alignment Workshop 2024. [\[Arxiv\]](#)

9. *"Chumor 1.0: A Truly Funny and Challenging Chinese Humor Understanding Dataset from Ruozhibai"*
Ruiqi He, Yushu He, Longju Bai, **Jiarui Liu**, Zhenjie Sun, Zenghao Tang, He Wang, Hanchen Xia, Naihao Deng
Preprint 2024. [\[Arxiv\]](#)
8. *"Implicit Personalization in Language Models: A Systematic Study"*
Zhijing Jin*, Nils Heil*, **Jiarui Liu***, Shehzaad Dhuliawala*, Yahang Qi*, Bernhard Schölkopf, Rada Mihalcea, Mrinmaya Sachan
EMNLP 2024 Findings. [\[Arxiv\]](#) [\[Code\]](#) [\[Video\]](#)
7. *"Synatra: Turning indirect knowledge into direct demonstrations for digital agents at scale"*
Tianyue Ou, Frank F Xu, Aman Madaan, **Jiarui Liu**, Robert Lo, Abishek Sridhar, Sudipta Sen-gupta, Dan Roth, Graham Neubig, Shuyan Zhou
NeurIPS 2024. [\[Arxiv\]](#)
6. *"Inducing Elasticity in Foundation Models: Post-Training Techniques for Adaptable Inference"*
Aashiq Muhamed, **Jiarui Liu**, Mona Diab, Virginia Smith
NeurIPS ENLSP Workshop 2024.
5. *"Automatic Generation of Model and Data Cards: A Step Towards Responsible AI"*
Jiarui Liu, Wenkai Li, Zhijing Jin, Mona Diab.
NAACL 2024 Oral. [\[ACL Anthology\]](#) [\[Arxiv\]](#) [\[Video\]](#) [\[Code\]](#)
4. *"Analyzing the Role of Semantic Representations in the Era of Large Language Models"*
Zhijing Jin*, Yuen Chen*, Fernando Gonzalez*, **Jiarui Liu**, Jiayi Zhang, Julian Michael, Bernhard Schölkopf, Mona Diab
NAACL 2024. [\[ACL Anthology\]](#) [\[Arxiv\]](#)
3. *"Can Large Language Models Infer Causation from Correlation?"*
Zhijing Jin*, **Jiarui Liu***, Zhiheng Lyu, Spencer Poff, Mrinmaya Sachan, Rada Mihalcea, Mona Diab, Bernhard Schölkopf
ICLR 2024. [\[Arxiv\]](#) [\[Code\]](#)
2. *"Bias Amplification Enhances Minority Group Performance"*
Gaotang Li*, **Jiarui Liu***, and Wei Hu.
TMLR 2023. [\[Arxiv\]](#) [\[Video\]](#)
1. *"Voices of Her: Analyzing Gender Differences in the AI Publication World"*
Yiwen Ding*, **Jiarui Liu***, Zhiheng Lyu*, Kun Zhang, Bernhard Schölkopf, Zhijing Jin[†], and Rada Mihalcea[†].
ACL 2025 NLP for Positive Impact Workshop. [\[Arxiv\]](#) [\[Code\]](#)

WORK EXPERIENCE

Meta

Research Intern

Redmond, US

May 2026 – Nov. 2026 (Expected)

Host: [Xin Luna Dong](#), [Renjie Tao](#), Tony Liao

- Accelerating autoresearch in machine learning through harness and RL for ML research ideation.

Amazon Stores Foundational AI, Rufus

Seattle, US

Applied Scientist Intern

May 2025 – Aug. 2025

Host: [Xian Li](#), [Jimmy Liu](#), [Xiaoman Pan](#)

- Formalized honest deductive reasoning as multi step tasks where models must either derive a conclusion strictly entailed by the premises or correctly abstain when information is insufficient.

Last updated in Jan. 2026

- Curated two graph based datasets for linear algebra and logical inference, introducing controlled unanswerable cases to evaluate model honesty under insufficient premises.
- Demonstrated that prompting and existing training methods, including GRPO, fail systematically due to unstable training dynamics, often collapsing when negative rewards dominate early learning.
- Proposed ANCHOR, a reinforcement learning method that injects ground truth trajectories into rollouts, stabilizing training and significantly improving honest reasoning performance.

Amazon AWS, Bedrock

New York, US

Applied Scientist Intern

Jun. 2024 – Aug. 2024

Host: *Yassine Benajiba, Renato Negrinho, Manuel Mager, Neha Anna John*

- Developed a sampling-based confidence calibration approach to mitigate object hallucination in Visual Question Answering (VQA) tasks.
- Introduced a novel atomic fact verification and refinement pipeline, reaching SOTA on 5 hallucination benchmarks and 2 multimodal models.
- Conducted experiments across both black-box and gray-box access to model logits and utilized visual contrastive decoding to further enhance calibration accuracy.

WarpEngine

Shanghai, China

Natural Language Processing R&D Intern, Host: *Hang Chu, Ming Liang*

May 2023 – Aug. 2023

- Constructed 14B language models with customized personalities by adopting different quantization, acceleration, and parameter efficient fine tuning methods (LoRA, P-Tuning).
- Built a pipeline for generating and cleaning customized dialogue data.
- Developed a comprehensive framework for evaluating open-domain dialogue capabilities of large language models.

TEACHING EXPERIENCE

Teaching Assistant for CMU's 11-768 AI Agents course in Fall 2026. Instructors: Graham Neubig and Daniel Fried.

Teaching Assistant for CMU's 11-830 Ethics, Safety, and Social Impact in NLP and LLMs in Spring 2026. Instructor: Maarten Sap.

PROFESSIONAL SERVICE

Conference Reviewer: NeurIPS 2026, ICML 2026, COLM 2026, ACL ARR 2026, ICLR 2026, NeurIPS 2025, ACL ARR 2025, ICLR 2025, CHI 2025

Workshop Organizer: PersonaNLP@NeurIPS Mexico, NLP for Positive Impacts 2025

TALKS

Invited talk at Nice-NLP on toward honest language models for deductive reasoning: Dec. 2025

Invited talk at Nice-NLP on automated model card generation and a multilingual AI terminology dataset: Jan. 2025

SKILLS

Programming: Python, C/C++, R, Java, Javascript, Go

Framework: PyTorch, Tensorflow, Transformers, Accelerate, DeepSpeed, BitsandBytes, NLTK, Scikit-Learn, XGBoost

Languages: Chinese (Native), English (Proficient, TOEFL 107 (S25 W30 R25 L27))